Notes on Local $\mathbb{F}_0$ and $\mathbb{F}_1$

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Abstract

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Contents

1. Local $\mathbb{F}_0$

1.1 Mirror curve

We write the mirror curve to $\mathbb{F}_0$ as the following embedding in $\mathbb{C}^* \times \mathbb{C}^* \ni (w, X)$

$$w + \frac{1}{w} + \frac{1}{R^2}\left(X + \frac{1}{X} + U\right) = 0. \tag{1.1}$$

We can write this curve as a quartic by introducing $Y = -R^2 X\left(w - \frac{1}{w}\right)$,

$$Y^2 = f_4(X) = (X^2 + UX + 1)^2 - 4X^2R^2. \tag{1.2}$$

The discriminant δ turns out to be (upto a numerical constant)

$$\delta = R^8\,\Delta(U,R), \quad \Delta(U,R) = 16R^8 - 8R^4\left(U^2+4\right) + \left(U^2-4\right)^2. \tag{1.3}$$

The equation $\Delta(U, R) = 0$ has four solutions

$$U_1 = 2 - 2R^2, \quad U_2 = -2 + 2R^2, \quad U_3 = -2 - 2R^2, \quad U_4 = 2 + 2R^2. \tag{1.4}$$

In the real $U - R$ slice, the condition is plotted in fig. 1.

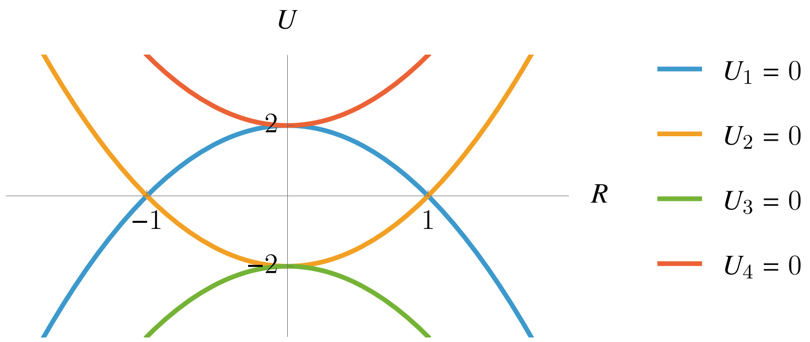


Figure 1: The locus $\Delta(U, R) = 0$

We can turn the quartic in eq. (1.2) to cubic form by the following rational transformation

$$X = \frac{ax+b}{cx+d}, \quad Y = \frac{ad-bc}{(cx+d)^2}y. \tag{1.5}$$

The curve now looks like

$$Y^2 = \frac{(ad-bc)^2}{(cx+d)^4}y^2 = f_4\left(\frac{ax+b}{cx+d}\right). \tag{1.6}$$

This implies,

$$(ad-bc)^2y^2 = f_4\left(\frac{ax+b}{cx+d}\right)(cx+d)^4 = c^4 f_4\left(\frac{a}{c}\right)x^4 + f_3(x), \tag{1.7}$$

where f_3 is a cubic polynomial in x whose coefficient of x^3 is directly proportional to $f_4'(a/c)$. It follows that if we set a/c to be a simple root of f_4 and $ad - bc = 1$ then we obtain the desired cubic form. We can then scale and translate so as to get the cubic in the usual Weierstrass form: $y^2 = 4x^3 - g_2 x - g_3$.

Performing this procedure on the quartic in eq. (1.2), one gets

$$\begin{aligned} g_2 &= \frac{1}{12}\left(16R^8 - 8R^4\left(U^2+2\right) + \left(U^2-4\right)^2\right), \\ g_3 &= \frac{1}{216}\left(4R^4 - U^2 + 4\right)\left(16R^8 - 8R^4\left(U^2+5\right) + \left(U^2-4\right)^2\right). \end{aligned} \tag{1.8}$$

We also find the j-invariant defined as $j = \frac{g_2^3}{g_2^3 - g_3^2}$,

$$j = \frac{\left(16R^8 - 8R^4\left(U^2+2\right) + \left(U^2-4\right)^2\right)^3}{R^8\left(16R^8 - 8R^4\left(U^2+4\right) + \left(U^2-4\right)^2\right)}. \tag{1.9}$$

1.2 Periods as double Laurent series

From the GLSM charges defining the toric geometry,

$$\begin{array}{ccc|cc} 0 & 0 & 1 & -2 & -2 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{array} \tag{1.10}$$

we see that the periods of the mirror curve are given by the double sum

$$\begin{aligned} \omega_1 &= \sum_{k,\ell\geq 0} \frac{(2k+2\ell)!}{(k!)^2(\ell!)^2} z^{k+\ell}\lambda^\ell = F_4\left(\tfrac{1}{2}, 1; 1, 1; 4z, 4z\lambda\right) \\ \omega_2 &= (2\log z + \log\lambda)\omega_1 - 2\sum_{k+\ell>0} \frac{(2k+2\ell)!}{(k!)^2(\ell!)^2}\left(H_k + H_\ell - 2H_{2k+2\ell}\right) z^{k+\ell}\lambda^\ell \end{aligned} \tag{1.11}$$

where F_4 is the Appell hypergeometric series, and $H_n = \sum_{k=1}^{n}(1/k)$ are the harmonic numbers. For $\lambda = 1$, we recover [1, A.39],

$$\omega_1 = {}_2F_1(\tfrac{1}{2},\tfrac{1}{2};1;16z), \quad \omega_2 = -2\pi\, {}_2F_1(\tfrac{1}{2},\tfrac{1}{2};1;1-16z), \tag{1.12}$$

Similarly, for $\lambda = -1$, we recover [1, A.46],

$$\omega_1 = {}_2F_1(\tfrac{1}{4},\tfrac{3}{4};1;-64z^2), \quad \omega_2 = -\pi\sqrt{2}\, {}_2F_1(\tfrac{1}{4},\tfrac{3}{4};1;1+64z^2), \tag{1.13}$$

The CY periods satisfy $z\partial_z\varpi_2 = \omega_1, z\partial_z\varpi_3 = \omega_2$. Hence they are obtained by integrating once with respect to z,

$$\begin{aligned}
\varpi_2 &= \log z + 2\sum_{k+\ell>0}\frac{(2k+2\ell-1)!}{(k!)^2(\ell!)^2}z^{k+\ell}\lambda^\ell \\
\varpi_3 &= -\log^2 z + (2\log z + \log\lambda)\,\varpi_2 - 2\sum_{k+\ell>0}\frac{(2k+2\ell)!}{(k!)^2(\ell!)^2(k+\ell)}\left(H_k + H_\ell - 2H_{2k+2\ell} + \frac{1}{k+\ell}\right)z^{k+\ell}\lambda^\ell
\end{aligned} \tag{1.14}$$

1.3 Kerr-Doran formula for the conifold period

In [2], a formula is provided for periods of nodal curves in toric ambient spaces. Here we check that this formula reproduces the result for the period at the conifold point derived in [A.84,A.107][1],

$$\begin{aligned}
T(\tau=0,m) &= \frac{2}{\pi^2}\left(\mathrm{Li}_2(\mathrm{i}e^{\mathrm{i}\pi m/2}) - \mathrm{Li}_2(-\mathrm{i}e^{\mathrm{i}\pi m/2})\right) \\
T(\tau=\tfrac{1}{2},m) &= \frac{2}{\pi^2}\left(\mathrm{Li}_2(e^{\mathrm{i}\pi m/2}) - \mathrm{Li}_2(-e^{\mathrm{i}\pi m/2})\right)
\end{aligned} \tag{1.15}$$

where $R = e^{\mathrm{i}\pi m/2}$, such that $R^4 = \lambda = e^{2\pi\mathrm{i}m}$.

1.3.1 Rational parametrization of the nodal curve

As shown in eq. (1.4), the curve in eq. (1.2) develops four distinct branches of nodes. These correspond to points in parameter space where the elliptic curve degenerates. Following the procedure outlined in [2], we rationally parametrize each nodal curve using a parameter z, such that the limits $z \to 0$ and $z \to \infty$ map to the node of the curve.

The quartic polynomial in eq. (1.2) takes the following forms along the four branches, each associated with a corresponding node:

$$\begin{aligned}
U_1 &= 2-2R^2, & f_4(X) &= (X+1)^2\left(X^2+\left(2-4R^2\right)X+1\right), & X_0 &= -1, \\
U_2 &= -2+2R^2, & f_4(X) &= (X-1)^2\left(X^2+\left(4R^2-2\right)X+1\right), & X_0 &= 1, \\
U_3 &= -2-2R^2, & f_4(X) &= (X-1)^2\left(X^2+\left(-4R^2-2\right)X+1\right), & X_0 &= 1, \\
U_4 &= 2+2R^2, & f_4(X) &= (X+1)^2\left(X^2+\left(4R^2+2\right)X+1\right), & X_0 &= -1.
\end{aligned} \tag{1.16}$$

To obtain a rational parametrization for the first branch (U_1), it suffices to find a rational expression for X such that $(X^2 + (2-4R^2)X + 1)$ becomes a perfect square. One convenient choice is

$$X = \frac{2\left(-2R^2 - t + 1\right)}{t^2 - 1}, \qquad \Rightarrow \qquad \left(X^2 + \left(2 - 4R^2\right)X + 1\right) = \left(\frac{\left(4R^2 - 2\right)t + t^2 + 1}{t^2 - 1}\right)^2. \tag{1.17}$$

The remaining parametrizations follow from the transformations $R^2 \to -R^2$ and $X \to -X$:

$$\begin{aligned}
X_1 &= \frac{2\left(-2R^2 - t + 1\right)}{t^2 - 1}, & Y_1 &= \frac{-16R^4 t + 4R^2(t-1)^3 + (t-1)^4}{\left(t^2-1\right)^2}, \\
X_2 &= -\frac{2\left(-2R^2 - t + 1\right)}{t^2 - 1}, & Y_2 &= \frac{-16R^4 t + 4R^2(t-1)^3 + (t-1)^4}{\left(t^2-1\right)^2}, \\
X_3 &= -\frac{2\left(2R^2 - t + 1\right)}{t^2 - 1}, & Y_3 &= \frac{-16R^4 t - 4R^2(t-1)^3 + (t-1)^4}{\left(t^2-1\right)^2}, \\
X_4 &= \frac{2\left(2R^2 - t + 1\right)}{t^2 - 1}, & Y_4 &= \frac{-16R^4 t - 4R^2(t-1)^3 + (t-1)^4}{\left(t^2-1\right)^2}.
\end{aligned} \tag{1.18}$$

Each pair (X_i, Y_i) satisfies eq. (1.2) for the corresponding value of U_i.

Returning to the original toric coordinates (X, w), the parametrizations become

$$\begin{aligned}
w_1 &= \frac{(t-1)\left(2R^2 + t - 1\right)}{2R^2(t+1)}, & w_2 &= -\frac{(t-1)\left(2R^2 + t - 1\right)}{2R^2(t+1)}, \\
w_3 &= \frac{(t-1)\left(2R^2 - t + 1\right)}{2R^2(t+1)}, & w_4 &= -\frac{(t-1)\left(2R^2 - t + 1\right)}{2R^2(t+1)}.
\end{aligned} \tag{1.19}$$

This yields a rational parametrization of the nodal curve in toric coordinates, though the node still lies at a finite distance rather than at 0 or ∞. To remedy this, we perform one final change of variables. For U_1, the node at $X_0 = -1$ corresponds to $t = 1 \pm 2R$, so we define

$$z = \frac{t - (1 - 2R)}{t - (1 + 2R)}, \tag{1.20}$$

similarly, for U_2. For U_3 and U_4, the node corresponds to $t = 1 \pm 2\mathrm{i}R$, so we define

$$z = \frac{t - (1 - 2\mathrm{i}R)}{t - (1 + 2\mathrm{i}R)}. \tag{1.21}$$

Therefore, for each branch we get X and w in the z-parametrization:

$$\begin{aligned}
X_1 &= -\frac{(z-1)(R(z-1)+z+1)}{(z+1)(Rz+R+z-1)}, & w_1 &= \frac{(z+1)(R(z-1)+z+1)}{(z-1)(Rz+R+z-1)}, \\
X_2 &= \frac{(z-1)(R(z-1)+z+1)}{(z+1)(Rz+R+z-1)}, & w_2 &= -\frac{(z+1)(R(z-1)+z+1)}{(z-1)(Rz+R+z-1)}, \\
X_3 &= \frac{(z-1)(R(z-1)+i(z+1))}{(z+1)(R(z+1)+i(z-1))}, & w_3 &= \frac{(z+1)(R(z-1)+i(z+1))}{(z-1)(R(z+1)+i(z-1))}, \\
X_4 &= -\frac{(z-1)(R(z-1)+i(z+1))}{(z+1)(R(z+1)+i(z-1))}, & w_4 &= -\frac{(z+1)(R(z-1)+i(z+1))}{(z-1)(R(z+1)+i(z-1))}.
\end{aligned} \tag{1.22}$$

1.3.2 Conifold period for a real curve

Applying the method of [2, Section 4.2], we find for the first branch the exponents and roots of the linear factors (with $i = 1$, $j = 1, \ldots, 4$):

$$\begin{aligned} (d_j) &= (1, 1, -1, -1), \qquad (\alpha_j) = \left(1, \frac{R-1}{R+1}, -1, \frac{1-R}{R+1}\right), \\ (e_j) &= (1, 1, -1, -1), \qquad (\beta_j) = \left(-1, \frac{R-1}{R+1}, 1, \frac{1-R}{R+1}\right). \end{aligned} \tag{1.23}$$

The (imaginary part of the) non-vanishing period at the conifold point is then

$$\mathcal{P}(R) = \frac{1}{2\pi} \sum_{j,k} d_j e_k D_2\left(\frac{\alpha_j}{\beta_k}\right), \tag{1.24}$$

where $D_2(z)$ is the Bloch–Wigner function,

$$D_2(z) = \mathrm{Im}(\mathrm{Li}_2(z)) + \arg(1-z)\, \log|z|, \tag{1.25}$$

and $\mathrm{Li}_2(z)$ denotes the ordinary dilogarithm [3].

We thus find, up to an overall sign,

$$\mathcal{P}(R) = \frac{1}{\pi}\left[-D_2\left(\frac{R+1}{1-R}\right) + D_2\left(\frac{R+1}{R-1}\right) + D_2\left(\frac{1-R}{R+1}\right) - D_2\left(\frac{R-1}{R+1}\right)\right]. \tag{1.26}$$

Using the Bloch–Wigner identities [3, Eq. (3)],

$$D_2(z) = D_2\left(1 - \frac{1}{z}\right) = D_2\left(\frac{1}{1-z}\right) = -D_2\left(\frac{1}{z}\right) = -D_2(1-z) = -D_2\left(\frac{-z}{1-z}\right), \tag{1.27}$$

we note that

$$D_2\left(\frac{1-R}{1+R}\right) = -D_2\left(\frac{R+1}{1-R}\right), \qquad D_2\left(\frac{R-1}{R+1}\right) = -D_2\left(\frac{R+1}{R-1}\right), \tag{1.28}$$

so that

$$\mathcal{P}(R) = \frac{2}{\pi}\left[D_2\left(\frac{R+1}{R-1}\right) - D_2\left(\frac{R+1}{1-R}\right)\right]. \tag{1.29}$$

We now use the five-term relation for the Bloch-Wigner,

$$D_2\left(\frac{1-x}{1-xy}\right) + D_2\left(\frac{1-y}{1-xy}\right) + D_2(1-xy) + D_2(x) + D_2(y) = 0, \tag{1.30}$$

with $x = R$, $y = -1$, to simplify the imaginary part of the period to the simple expression

$$\mathcal{P}(R) = \frac{2}{\pi}\left[D_2(-R) - D_2(R)\right]. \tag{1.31}$$

Since $D_2(\bar{z}) = -D_2(z)$, it follows that

$$\mathcal{P}(R) = 0 \qquad \text{for real } R. \tag{1.32}$$

Repeating the analysis for U_2 yields identical exponents and roots, and thus the same result. For U_3 and U_4, the change of variables is $t = \frac{-2\mathrm{i}R(z+1)+z-1}{z-1}$, giving

$$\begin{aligned} (d_j) &= (1,1,-1,-1), \qquad (\alpha_j) = \left(1, \frac{R-\mathrm{i}}{R+\mathrm{i}}, -1, \frac{\mathrm{i}-R}{R+\mathrm{i}}\right), \\ (e_j) &= (1,1,-1,-1), \qquad (\beta_j) = \left(-1, \frac{R-\mathrm{i}}{R+\mathrm{i}}, 1, \frac{\mathrm{i}-R}{R+\mathrm{i}}\right). \end{aligned} \tag{1.33}$$

Hence,

$$\begin{aligned} \mathcal{P}(R) &= \frac{1}{\pi}\left[-D_2\left(\frac{R+i}{-R+i}\right) + D_2\left(\frac{R+i}{R-i}\right) + D_2\left(\frac{-R+i}{R+i}\right) - D_2\left(\frac{R-i}{R+i}\right)\right] \\ &= \frac{2}{\pi}\left[D_2\left(\frac{-R+i}{R+i}\right) - D_2\left(\frac{R-i}{R+i}\right)\right]. \end{aligned} \tag{1.34}$$

Applying the five term relation gives us

$$\tilde{\mathcal{P}}(R) = \frac{2}{\pi}\left[D_2(\mathrm{i}R) - D_2(-\mathrm{i}R)\right] \tag{1.35}$$

And this time we see that But the formula should not apply

$$\mathcal{P}(R) = 0 \qquad \text{for imaginary } R. \tag{1.36}$$

1.3.3 Conifold period for general moduli

In fact, apart from the imaginary part of the period, [2, eq. (4.12)] also gives a formula for the full non-vanishing period at the conifold points, even when the curve is not real:

$$\begin{aligned} Q &= \frac{1}{2\pi}\sum_{j,k} d_j e_k \left[\mathrm{Li}_2\left(\frac{\alpha_j}{\beta_k}\right) + (\log\alpha_j - \log\beta_k)\log\left(1 - \frac{\alpha_j}{\beta_k}\right)\right] \\ &\quad - \frac{1}{2\pi}\log y(0)(\log x(0) - \log x(\infty)) + \log y(0)\sum_j d_j \log\alpha_j \\ &\quad - \log x(\infty)\sum_k e_k \log\beta_k. \end{aligned} \tag{1.37}$$

where as before, $x(z)$ and $y(z)$ are rational parametrizations of the nodal curve in the original toric coordinates and the node corresponds to $z \to 0, \infty$. From now on, we work with the branches U_1 and U_2 but all the results can be generalized to U_3 and U_4 by taking $R \to \mathrm{i}R$. Evaluating eq. (1.37) for the parametrization in eq. (1.22), we find the following after assuming $\operatorname{Im} R > 0$ and simplifying logs

$$\begin{aligned} Q &= \frac{1}{\pi}\left(\mathrm{Li}_2\left(\frac{-R-1}{R-1}\right) - \mathrm{Li}_2\left(\frac{R+1}{R-1}\right) - \mathrm{Li}_2\left(\frac{1-R}{R+1}\right) + \mathrm{Li}_2\left(\frac{R-1}{R+1}\right)\right) \\ &\quad - \frac{2}{\pi}\log(R)\log(R-1) + i\log(R-1) + 2i\log(R) \\ &\quad + \frac{2}{\pi}\log(R)\log(R+1) - i\log(R+1). \end{aligned} \tag{1.38}$$

This expression can be simplified further thanks to various identities of the dilogarithm. We use in particular the following identities [3]

$$\mathrm{Li}_2\left(\frac{1}{z}\right) = -\mathrm{Li}_2(z) - \frac{\pi^2}{6} - \frac{1}{2}\log(-z)^2, \tag{1.39}$$

$$\mathrm{Li}_2\left(1-z\right) = -\mathrm{Li}_2(z) + \frac{\pi^2}{6} - \log(z)\log(1-z). \tag{1.40}$$

We also have the so-called *five term relation*,

$$\begin{aligned}&\mathrm{Li}_2(x) + \mathrm{Li}_2(y) + \mathrm{Li}_2\left(\frac{x-1}{xy-1}\right) + \mathrm{Li}_2\left(\frac{y-1}{xy-1}\right) + \mathrm{Li}_2(1-xy)\\ &= \frac{\pi^2}{2} - \log(1-x)\log(x) - \log(1-y)\log(y) - \log\left(\frac{1-x}{1-xy}\right)\log\left(\frac{1-y}{1-xy}\right),\end{aligned} \tag{1.41}$$

which only holds near the viscinity of $x, y \in (0,1)$ due to branch cuts. However, it is easy to find a generalized version of this identify for the domain we are interested in by starting with the identity in it's twice differentiated form (where it reduces to a trivial algebraic identity) and integrating twice and fixing the integration constants.

Using the inversion identity eq. (1.39), we get

$$Q = -\frac{\pi}{2} - \frac{2}{\pi}\left[\mathrm{Li}_2\left(\frac{1-R}{R+1}\right) - \mathrm{Li}_2\left(\frac{R-1}{R+1}\right) - \log\left(\frac{1-R}{R+1}\right)\log(R)\right]. \tag{1.42}$$

We now invoke the five term relation with $(x, y) = (R, -1)$ and correct it with the procedure explained. After applying the reflection identity to some of the arguments eq. (1.40) and noting that $\mathrm{Li}_2(-1) = \frac{-\pi^2}{12}$, the identity reads simply

$$-\mathrm{Li}_2(-R) + \mathrm{Li}_2(R) + \mathrm{Li}_2\left(\frac{1-R}{R+1}\right) - \mathrm{Li}_2\left(\frac{R-1}{R+1}\right) = \frac{\pi^2}{4} - \log(R)\log\left(\frac{1-R}{R+1}\right). \tag{1.43}$$

Applying this to Q, we get our final expression

$$Q = -\pi - \frac{2}{\pi}\left[\mathrm{Li}_2(-R) - \mathrm{Li}_2(R)\right]. \tag{1.44}$$

2. Local $\mathbb{F}_1$

2.1 Nagell's algorithm for the mirror curve

We start from the equation of the local mirror of $\mathbb{F}_1$ defined as the elliptic curve,

$$x + y + \frac{1}{xy} + \frac{m}{x} = \frac{1}{u}, \tag{2.1}$$

where m and u are a priori free complex parameters. After multiplying by xy, we have

$$f(x, y) = x^2y + xy^2 - \frac{xy}{u} + my + 1 = 0. \tag{2.2}$$

We can eliminate the constant term by shifting $y \to y - \frac{1}{m}$,

$$f(x,y) = x^2y + xy^2 - \frac{x^2}{m} - \frac{(2u+m)xy}{um} + my + \frac{(m+u)x}{m^2u}. \tag{2.3}$$

We perform a series of affine transformations on x and y known as Nagell's algorithm as described in [4] to bring the curve into Weierstrass form. Note that in order for the curve to be elliptic, m and u cannot vanish simultaneously. Since we would be interested in the $m \to 0$ limit (Local $\mathbb{P}^2$), we switch $x \leftrightarrow y$ and assume generically that $m + u \neq 0$. We thus have

$$f(x,y) = x^2y + xy^2 - \frac{y^2}{m} - \frac{(2u+m)xy}{um} + mx + \frac{(m+u)y}{m^2u}. \tag{2.4}$$

Let us homogenize by writing $x = X/Z$, $y = X/Z$ and clearing denominators. We get,

$$F(X,Y,Z) = F_3(X,Y) + F_2(X,Y)Z + F_1(X,Y)Z^2. \tag{2.5}$$

where

$$F_3 = X^2Y + XY^2, \tag{2.6}$$
$$F_2 = -\frac{(2u+m)}{um}XY - \frac{Y^2}{m}, \tag{2.7}$$
$$F_1 = mX + \frac{(m+u)}{m^2u}Y. \tag{2.8}$$

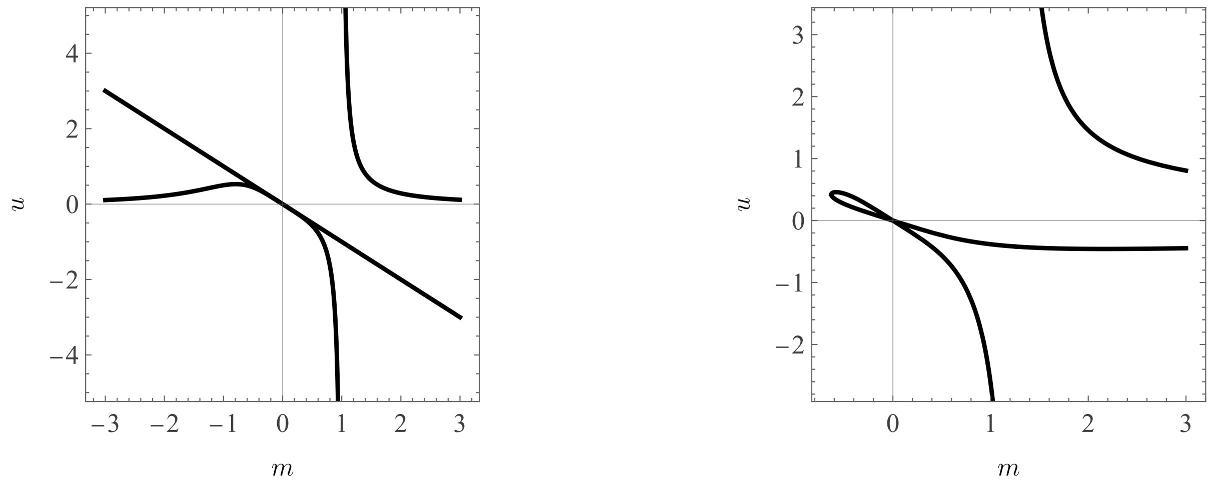
Note that the point $P = [0,0,1]$ is rational by construction (after eliminating the constant term). The tangent at P is given by $F_1 = mX + \frac{(m+u)}{m^2u}Y = 0$ and meets the curve at another point Q. We also define $e_i = F_i\left(\frac{(m+u)}{m^2u}, -m\right)$ ($e_1 = 0$ by definition) —

$$e_2 = \frac{2}{m^2} + \frac{3}{mu} - m + \frac{1}{u^2}, \tag{2.9}$$
$$e_3 = \frac{(m+u)\left(u - \frac{m+u}{m^3}\right)}{u^2}. \tag{2.10}$$

The coordinates of Q are given by

$$\begin{aligned} Q = \Big[&\frac{(m+u)\left(m^3u^2 - m^2 - 3mu - 2u^2\right)}{m^4u^3}, \\ &-m^2 + \frac{m}{u^2} + \frac{2}{m} + \frac{3}{u}, \quad \frac{(m+u)\left(u - \frac{m+u}{m^3}\right)}{u^2} \Big]. \end{aligned} \tag{2.11}$$

For the curve to be elliptic, e_2 and e_3 cannot vanish simultaneously which is true provided m and u don't vanish simultaneously. On the real section of the $m - u$ plane, these conditions are plotted in fig. 2.



(a) $e_3 = 0$ (b) $e_2 = 0$

Figure 2: The loci where e_2, e_3 vanish.

Assuming generically that $e_3 \neq 0$, we make the coordinate change

$$X = X' + \frac{mu\left(\frac{2}{m^2} + \frac{3}{mu} - m + \frac{1}{u^2}\right)}{m + u - m^3 u} Z' \tag{2.12}$$

$$Y = Y' - \frac{m^2\left(m^3\left(-u^2\right) + m^2 + 3mu + 2u^2\right)}{(m+u)\left(m + u - m^3 u\right)} Z' \tag{2.13}$$

$$Z = Z' \tag{2.14}$$

which shifts Q to the origin, $[X', Y', Z'] = [0, 0, 1]$. We now shift back to affine coordinates $x' = X'/Z'$, $y' = Y'/Z'$. The curve can now be written as

$$f'(x', y') = f_3'(x', y') + f_2'(x', y') + f_1'(x', y') \tag{2.15}$$

where

$$f_3' = x'^2 y' + x' y'^2 \tag{2.16}$$

$$\begin{aligned} f_2' &= \frac{(m+u)^2}{mu\left(m + u - m^3 u\right)} y'^2 + \left(\frac{2\left(m + u - m^3 u\right)}{m(m+u)} + \frac{1}{u}\right) x' y' \\ &\quad + \frac{m^2\left(-\left(m^3 - 2\right)u^2 + m^2 + 3mu\right)}{(m+u)\left(\left(m^3 - 1\right)u - m\right)} x'^2 \end{aligned} \tag{2.17}$$

$$\begin{aligned} f_1' &= \frac{(m+u)\left(2m^6u^2 - m^5 - 5m^4u - 4m^3u^2 + m^2 + 2mu + u^2\right)}{m^2 u\left(m + u - m^3 u\right)^2} y' \\ &\quad + \frac{m}{u(m+u)^2\left(m + u - m^3 u\right)^2}\Big(-m^5 - 6m^4 u + \left(m^3 - 15\right)m^3u^2 \\ &\qquad - 3\left(m^6 - 3m^3 + 4\right)mu^4 + \left(6m^3 - 19\right)m^2u^3 \\ &\qquad + \left(m^9 - 3m^6 + 4m^3 - 3\right)u^5\Big)x' \end{aligned} \tag{2.18}$$

We define $\phi_i = f_i'(1,t)$ and

$$\begin{aligned}
\delta &= \phi_2^2 - 4\phi_1\phi_3 \\
&= \frac{(m+u)^4}{m^2u^2\left(m+u-m^3u\right)^2}t^4 \\
&\quad - \frac{2(m+u)\left(-\left(5m^3+1\right)u^2 - m\left(m^3+2\right)u - m^2 + 2m^2\left(m^3-2\right)u^3\right)}{mu^2\left(m+u-m^3u\right)^2}t^3 \\
&\quad + \frac{1}{u^2\left(m+u-m^3u\right)^2}\Big(2m\left(2m^3+1\right)u + m^2 + 2m\left(-2m^6+m^3+4\right)u^4 \\
&\qquad + \left(m^6+18m^3+1\right)u^2 + 2m^2\left(m^3+11\right)u^3\Big)t^2 \\
&\quad + \frac{2m}{u(m+u)\left(m+u-m^3u\right)^2}\Big(m^4 + \left(m^3+4\right)m^3u + 2\left(-m^6+m^3+1\right)u^4 \\
&\qquad + \left(-3m^6+7m^3+6\right)mu^3 + \left(6m^3+7\right)m^2u^2\Big)t \\
&\quad + \frac{m^4\left(-\left(m^3-2\right)u^2+m^2+3mu\right)^2}{(m+u)^2\left(m+u-m^3u\right)^2}
\end{aligned} \tag{2.19}$$

Note that $\delta = 0$ whenever t is the slope of a tangent to the curve that passes through Q, in particular for $t = -\frac{m^3u}{m+u}$ (can be checked). We write $t = \frac{1}{\tau} - \frac{m^3u}{m+u}$ and define $\rho = \tau^4\delta$ which is a cubic in τ

$$\begin{aligned}
\rho &= \frac{4m\left(m+u-m^3u\right)}{m+u}\tau^3 + \left(8m+\frac{1}{u^2}\right)\tau^2 + \frac{2(m+u)\left(2m^2u^2+m+u\right)}{mu^2\left(m+u-m^3u\right)}\tau \\
&\quad + \frac{(m+u)^4}{m^2u^2\left(m+u-m^3u\right)^2}.
\end{aligned} \tag{2.20}$$

Applying the final transformation,

$$\rho = \frac{(m+u)^2}{m^2\left(m+u-m^3u\right)^2}V^2, \tag{2.21}$$

$$\tau = \frac{(m+u)}{m\left(m+u-m^3u\right)}\left(U - \frac{1}{12}\left(8m+\frac{1}{u^2}\right)\right), \tag{2.22}$$

we obtain the final Weierstrass form,

$$V^2 = 4U^3 - g_2U + g_3, \tag{2.23}$$

with

$$g_2 = -\frac{4}{3}\left(-16m^2 + \frac{8m}{u^2} - \frac{1}{u^4} + \frac{24}{u}\right), \tag{2.24}$$

$$g_3 = -\frac{8}{27}\left(-64m^3 + \frac{48m^2}{u^2} - \frac{12m}{u^4} + \frac{144m}{u} + \frac{1}{u^6} - \frac{36}{u^3} + 216\right). \tag{2.25}$$

2.2 Periods as double Laurent series

From the GLSM charges defining the toric geometry,

$$\begin{array}{ccc|cc} 0 & 0 & 1 & -2 & -1 \\ 1 & 0 & 1 & 1 & 0 \\ -1 & -1 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 & 1 \end{array} \tag{2.26}$$

we see that the A-period of the mirror curve is given by the double sum

$$\begin{aligned} \omega_1 &= \sum_{k,\ell\geq 0} \frac{(k+2\ell)!}{(k!)^2(l!)(\ell-k)!}(-u)^{-k-2\ell}\lambda^{\ell-k} \\ &= \sum_{k,\ell\geq 0} \frac{(3k+2\ell)!}{(k!)^2(k+\ell)!(\ell!)}(-u)^{-3k-2\ell}\lambda^{\ell} \end{aligned} \tag{2.27}$$

where $z_1 = \frac{\lambda}{u^2}$, $z_2 = \frac{1}{u\lambda}$ are the standard complex structure parameters? related to Marcos' parameters by $u_{\text{here}} = -\frac{1}{u_{\text{Marcos}}}$, $\lambda = m$?
where in the second line we shifted $\ell \mapsto \ell + k$, observing that terms with $k > \ell$ do not contribute in the sum in the first line. Indeed, we check that ω_1 is annihilated by the second order operator

$$\begin{aligned} \mathcal{D} =& (8\lambda u - 9)\left(16\lambda^3 + \lambda u^4 - u^3 - 8\lambda^2u^2 + 36\lambda u - 27\right)\theta_u^2 \\ &+ (-729 + 1512u\lambda - u^4\lambda - 936u^2\lambda^2 + 432\lambda^3 + 128u^3\lambda^3 - 512u\lambda^4)\theta_u \\ &- 2(-243 + 486u\lambda - 282u^2\lambda^2 + 144\lambda^3 + 32u^3\lambda^3 - 192u\lambda^4) \end{aligned} \tag{2.28}$$

such that $\mathcal{D}_3 = u^{-3}\mathcal{D}\partial_u$. Using the Frobenius method, we find two logarithmic solutions

$$\begin{aligned} \omega_2 =& (-\log u - \log\lambda)\omega_1 + 2\sum_{-2\ell<k\leq\ell} \frac{(k+2\ell)!}{(k!)^2(k)!(\ell-k)!}\left(H_{2\ell+k} + H_{\ell-k} - 2H_k\right)(-u)^{-k-2\ell}\lambda^{\ell-k} \\ &+ \sum_{0\leq\ell<k}(-1)^\ell \frac{\Gamma(k-\ell)(k+2\ell)!}{\ell!\,(k!)^2}u^{-k-2\ell}\lambda^\ell \\ \omega_2' =& (-2\log u + \log\lambda)\omega_1 + 2\sum_{-2\ell<k\leq\ell} \frac{(k+2\ell)!}{(k!)^2(k)!(\ell-k)!}\left(2H_{2\ell+k} - H_{\ell-k} - H_\ell\right)(-u)^{-k-2\ell}\lambda^{\ell-k} \\ &- \sum_{0\leq\ell<k}(-1)^\ell \frac{\Gamma(k-\ell)(k+2\ell)!}{\ell!\,(k!)^2}u^{-k-2\ell}\lambda^\ell \end{aligned} \tag{2.29}$$

where $H_n = \sum_{k=1}^n (1/k)$ are the harmonic numbers. In fact, only the linear combination $2\omega_2 + 3\omega_2'$ is annihilated by $\mathcal{D}$, while ω_2 and ω_2' satisfy it with a source term proportional to $9\lambda u^4 + 96\lambda^3u^3$.

The CY periods satisfy $u\partial_u\varpi_2 = \omega_1, u\partial_u\varpi_3 = \frac{1}{4}(2\omega_2 + 3\omega_2')$. Hence they are obtained by integrating once with respect to u,

$$
\begin{aligned}
\varpi_2 &= -\log u + \sum_{3k+2\ell>0} \frac{(3k+2\ell-1)!}{(k!)^2(\ell!)(k+\ell)!}(-u)^{-3k-2\ell}\lambda^\ell \\
\varpi_3 &= \log^2 u + (2\log u - \frac{1}{4}\log\lambda)\varpi_2 + \frac{1}{8}\log^2\lambda \\
&\quad - 2\sum_{-2\ell<k\le\ell} \frac{(k+2\ell-1)!}{(k!)^2(k)!(\ell-k)!}\left(2H_{2\ell+k} - H_{\ell-k} - H_\ell\right)(-u)^{-k-2\ell}\lambda^{\ell-k} \\
&\quad + \sum_{0\le\ell<k}(-1)^\ell \frac{\Gamma(k-\ell)(k+2\ell-1)!}{\ell!\,(k!)^2}u^{-k-2\ell}\lambda^\ell - 2\sum_{k+2\ell>0}\frac{(k+2\ell-1)!}{(k+2\ell)(k!)^2(k)!(\ell-k)!}u^{-k-2\ell}\lambda^\ell
\end{aligned}
\tag{2.30}
$$

To be checked numerically

2.3 Convergence domain of Frobenius solution as a function of m

Consider the series

$$
f(u,m) \;=\; \log u \;+\; \sum_{n\ge 2}\sum_{3k+2\ell=n} \frac{(n-1)!}{(k+\ell)!\,(k!)^2\,\ell!}\, m^\ell\, u^n\,. \tag{2.31}
$$

For fixed complex parameter m, this is a series in the complex variable u. We are interested in the radius of convergence $R(m)$ of the power–series part around $u = 0$.

The logarithmic term $\log u$ contributes a singularity at $u = 0$, but it does not affect the radius of convergence of the power–series part. Thus we focus on

$$
g(u,m) := \sum_{n\ge 2} a_n(m)\, u^n, \qquad a_n(m) := \sum_{3k+2\ell=n} \frac{(n-1)!}{(k+\ell)!\,(k!)^2\,\ell!}\, m^\ell\,. \tag{2.32}
$$

The radius of convergence for fixed m is

$$
\frac{1}{R(m)} = \limsup_{n\to\infty} |a_n(m)|^{1/n}. \tag{2.33}
$$

2.3.1 Rewriting as a double series

It is convenient to write the power series directly as a sum over (k,ℓ):

$$
g(u,m) \;=\; \sum_{k,\ell\ge 0} \frac{(3k+2\ell-1)!}{(k+\ell)!\,(k!)^2\,\ell!}\, m^\ell\, u^{3k+2\ell}. \tag{2.34}
$$

For fixed m, u, the general term in absolute value is

$$
T_{k,\ell}(R,M) \;:=\; \left|\frac{(3k+2\ell-1)!}{(k+\ell)!\,(k!)^2\,\ell!}\right| R^{3k+2\ell}\, M^\ell, \qquad R = |u|, \quad M = |m|. \tag{2.35}
$$

The radius $R(m)$ is determined by the exponential growth of the coefficients in $n := 3k + 2\ell$, i.e. by $\limsup |a_n(m)|^{1/n}$.

2.3.2 Saddle–point analysis via Stirling

For large k, ℓ with fixed ratio, we approximate factorials using Stirling's formula:

$$n! \sim n^n e^{-n} \sqrt{2\pi n}. \tag{2.36}$$

Introduce the scaling

$$k = \alpha n, \qquad \ell = \beta n, \qquad n := 3k + 2\ell, \tag{2.37}$$

so that

$$3\alpha + 2\beta = 1, \qquad \alpha, \beta \geq 0. \tag{2.38}$$

A single term in the inner sum defining $a_n(m)$ then behaves like

$$\log \left| \frac{(n-1)!}{(k+\ell)!\,(k!)^2\,\ell!}\, m^\ell \right| \sim n\, \phi(\alpha, \beta; M), \tag{2.39}$$

for some function ϕ that can be extracted from the Stirling approximation. Using the constraint $3\alpha + 2\beta = 1$, we can solve

$$\alpha = \frac{1 - 2\beta}{3}, \qquad 0 \leq \beta \leq \frac{1}{2}. \tag{2.40}$$

In terms of β alone, one finds

$$\phi(\beta; M) \;=\; \log\left(\frac{M^\beta}{\alpha^{2\alpha}\, \beta^\beta\, (\alpha+\beta)^{\alpha+\beta}} \right), \qquad \alpha = \frac{1-2\beta}{3}, \tag{2.41}$$

up to subleading (subexponential in n) terms.

For large n, the sum over ℓ (equivalently over β) is dominated by the value of β that maximizes $\phi(\beta; M)$ for fixed $M = |m|$. Hence

$$\limsup_{n\to\infty} |a_n(m)|^{1/n} = \exp\left(\sup_{0 \leq \beta \leq 1/2} \phi(\beta; M) \right). \tag{2.42}$$

Denote

$$\log \Lambda(M) := \sup_{0 \leq \beta \leq 1/2} \phi(\beta; M). \tag{2.43}$$

Then the radius of convergence is

$$R(m) = \frac{1}{\Lambda(|m|)}. \tag{2.44}$$

2.3.3 Stationary point and parametric description

The dominant contribution comes from a saddle point $\beta_*(M)$ where

$$\frac{d\phi}{d\beta}(\beta_*(M); M) = 0. \tag{2.45}$$

A straightforward differentiation and simplification give

$$\frac{d\phi}{d\beta} = \log\left(\frac{M}{3\beta(\beta+1)^{1/3}}\right) + \frac{4}{3}\log(1-2\beta). \tag{2.46}$$

The saddle condition thus reads

$$\log\left(\frac{M}{3\beta(\beta+1)^{1/3}}\right) = -\frac{4}{3}\log(1-2\beta), \tag{2.47}$$

or equivalently

$$M = 3\beta(\beta+1)^{1/3}(1-2\beta)^{-4/3}. \tag{2.48}$$

For each $M = |m| > 0$, there is (under suitable regularity assumptions) a unique $\beta_*(M) \in (0, 1/2)$ solving this equation.

Evaluating the growth factor $\Lambda(M)$ at the saddle, one finds that the corresponding exponential growth rate can be written as

$$\Lambda(M) = E(\beta_*(M)), \tag{2.49}$$

where

$$E(\beta) = 3\,\frac{(1-2\beta)^{-2/3}}{(1+\beta)^{1/3}}. \tag{2.50}$$

Therefore, the radius of convergence is

$$R(m) = \frac{1}{\Lambda(|m|)} = \frac{1}{3}\,(1-2\beta_*)^{2/3}(1+\beta_*)^{1/3}, \tag{2.51}$$

with $\beta_* = \beta_*(|m|)$ determined implicitly by (2.48).

In other words, we can give a parametric description of the boundary of the convergence domain in the $(|u|, |m|)$–plane. Introducing a real parameter $\beta \in (0, 1/2)$, we set

$$\boxed{\begin{aligned} M(\beta) &= 3\beta\,(1+\beta)^{1/3}(1-2\beta)^{-4/3}, \\ R(\beta) &= \frac{1}{3}\,(1-2\beta)^{2/3}(1+\beta)^{1/3}, \end{aligned}} \tag{2.52}$$

so that for fixed m with $M = |m|$, one determines the corresponding β via $M(\beta) = |m|$, and then the radius of convergence is $R(m) = R(\beta)$.

2.3.4 Special limits

The case $m = 0$. Taking the limit $M \to 0$ corresponds to $\beta \to 0$. From $M(\beta) \sim 3\beta$ as $\beta \to 0$, we have $\beta \sim M/3$, while

$$R(\beta) \xrightarrow[\beta\to 0]{} \frac{1}{3}. \tag{2.53}$$

Thus

$$R(0) = \frac{1}{3}. \tag{2.54}$$

Large $|m|$. For $|m| \to \infty$, the parameter β approaches $1/2$ from below. From the parametric form one can extract the asymptotic behaviour

$$R(m) \sim C\,|m|^{-1/2}, \qquad |m| \to \infty, \tag{2.55}$$

for some explicit constant $C > 0$ (which can be worked out from the leading behaviour of $M(\beta)$ and $R(\beta)$ near $\beta = 1/2$).

2.3.5 Summary

For each fixed value of the parameter $m \in \mathbb{C}$, the power–series part of $f(u, m)$ in u has a finite radius of convergence $R(m)$ given implicitly by a saddle–point analysis:

- Introduce $M = |m|$ and a parameter $\beta \in (0, 1/2)$.
- The pair (M, R) is parametrized by

$$M(\beta) = 3\beta\,(1+\beta)^{1/3}(1-2\beta)^{-4/3}, \qquad R(\beta) = \frac{1}{3}\,(1-2\beta)^{2/3}(1+\beta)^{1/3}. \tag{2.56}$$

- Given m, solve $M(\beta) = |m|$ for β and then $R(m) = R(\beta)$.

The logarithmic term $\log u$ does not affect the radius of convergence of the power–series part, though it introduces a singularity at $u = 0$ itself.

2.4 Rational change of coordinates and the Kerr-Doran prescription

Nagell's algorithm provides a systematic procedure to convert the mirror curve of local $\mathbb{F}_1$ into Weierstrass form, but it does so at the price of introducing a transcendental change of variables. A significantly simpler rational transformation, originally presented in [5], is given by

$$x = \frac{36u^2(m+3u) + 6X + \sqrt{2}\,Y - 9}{6u\left(12mu^2 + 2X - 3\right)}, \qquad y = \frac{36u^2}{-12mu^2 - 2X + 3}. \tag{2.57}$$

In these coordinates the mirror curve eq. (2.1) assumes Weierstrass form,

$$Y^2 = 4X^3 - g_2 X - g_3, \tag{2.58}$$

with

$$\begin{aligned} g_2 &= 27\left(16m^2u^4 - 8mu^2 - 24u^3 + 1\right), \\ g_3 &= 27\left(64m^3u^6 - 48m^2u^4 - 144mu^5 + 12mu^2 - 216u^6 + 36u^3 - 1\right). \end{aligned} \tag{2.59}$$

These expressions agree with eq. (2.24) up to an overall normalization, which can be absorbed by rescaling X and Y. The discriminant takes the form

$$\Delta \propto u^8\,\delta(u,m), \qquad \delta(u,m) = 16m^3u^4 - 8m^2u^2 - 36mu^3 + m - 27u^4 + u, \tag{2.60}$$

showing that for fixed m the curve generically develops four nodal degenerations corresponding to the roots of $\delta(u,m) = 0$. Among these solutions we single out the branch $u_*(m)$ of smallest absolute value; this root is the one governing, for example, the convergence of the large-radius expansion of the periods, and will therefore play a distinguished role in the analysis.

To treat all nodal branches uniformly we follow the formalism of [5]. At the nodal locus the curve can be written as

$$Y^2 = 4(X - X_0)^2(X - X_1). \tag{2.61}$$

Eliminating the X^2 term enforces $X_1 = -2X_0$. Matching coefficients with the Weierstrass form then yields

$$X_0 = -\frac{3g_3}{2g_2}, \tag{2.62}$$

evaluated on the discriminant locus $\delta(u,m) = 0$. A convenient rational parametrization of eq. (2.61) is

$$X = \frac{1}{2}\left(t^2 - 4X_0\right), \qquad Y = \frac{t\left(t^2 - 6X_0\right)}{\sqrt{2}}. \tag{2.63}$$

Expressed back in the original toric coordinates, this leads to

$$\begin{aligned} x &= \frac{36u^2(m+3u) + 3(t^2 - 4X_0) + t(t^2 - 6X_0) - 9}{6u\left(12mu^2 + t^2 - 4X_0 - 3\right)}, \\ y &= \frac{36u^2}{-12mu^2 - t^2 + 4X_0 + 3}. \end{aligned} \tag{2.64}$$

Introducing the notation

$$t_1^2 = -12mu^2 + 4X_0 + 3, \tag{2.65}$$

the denominators become $t^2 - t_1^2 = (t - t_1)(t + t_1)$. On the discriminant locus one may verify that the numerator of x factorizes as

$$36u^2(m+3u) + 3(t^2 - 4X_0) + t(t^2 - 6X_0) - 9 = (t - t_1)^2(t + 2t_1 + 3), \tag{2.66}$$

which leads to the one-parameter family

$$x = \frac{(t - t_1)(t + 2t_1 + 3)}{6u(t + t_1)}, \qquad y = -\frac{36u^2}{t^2 - t_1^2}. \tag{2.67}$$

To apply the Kerr–Doran prescription we shift the nodes to 0 and ∞. From eq. (2.63) the node occurs at $t = \pm\sqrt{6X_0}$, leading to the coordinate

$$z = \frac{t - \sqrt{6X_0}}{t + \sqrt{6X_0}}. \tag{2.68}$$

In terms of z, the parametrization eq. (2.67) takes the form

$$x = \alpha \, \frac{(z-a)(z-b)}{(z-c)(z-1)}, \qquad y = \beta \, \frac{(z-1)^2}{(z-b)(z-c)}, \tag{2.69}$$

where

$$\begin{gathered} b = \frac{t_1 - \sqrt{6X_0}}{t_1 + \sqrt{6X_0}}, \qquad a = \frac{3 + 2t_1 + \sqrt{6X_0}}{3 + 2t_1 - \sqrt{6X_0}} = \frac{1}{b^2}, \qquad c = \frac{t_1 + \sqrt{6X_0}}{t_1 - \sqrt{6X_0}} = \frac{1}{b}, \\ \alpha = \frac{t_1 + 3}{6u} = \left(\sqrt{b} + \frac{1}{\sqrt{b}}\right)^{-\frac{4}{3}}, \qquad \beta = -\frac{18u^2}{t_1^2 + 3t_1} = \left(\sqrt{b} + \frac{1}{\sqrt{b}}\right)^{\frac{2}{3}}. \end{gathered} \tag{2.70}$$

Using the general expression for the imaginary part of the period,

$$\mathcal{P} = \frac{1}{2\pi} \sum_{j,k} d_j e_k \, D_2\!\left(\frac{\alpha_j}{\beta_k}\right), \tag{2.71}$$

one finds

$$\mathcal{P}(b) = \frac{1}{2\pi} \left[D_2(b^3) - 4D_2(b^2) + 5D_2(b)\right]. \tag{2.72}$$

The full non-vanishing period follows from

$$\begin{aligned} Q = {} & \frac{1}{2\pi} \sum_{j,k} d_j e_k \left[\mathrm{Li}_2\!\left(\frac{\alpha_j}{\beta_k}\right) + (\log\alpha_j - \log\beta_k)\, \log\!\left(1 - \frac{\alpha_j}{\beta_k}\right)\right] \\ & - \frac{1}{2\pi} \log y(0)\, \left(\log x(0) - \log x(\infty)\right) + \log y(0) \sum_j d_j \log\alpha_j - \log x(\infty) \sum_k e_k \log\beta_k, \end{aligned} \tag{2.73}$$

giving the explicit expression

$$\begin{aligned} Q(b) = {} & \frac{1}{2\pi} \left[\mathrm{Li}_2(b^3) - 4\,\mathrm{Li}_2(b^2) + 5\,\mathrm{Li}_2(b)\right] \\ & + \frac{1}{4\pi} \log b \left(6\log\!\left(b^2 + b + 1\right) + \log b - 16\log(b+1) - 4\mathrm{i}\pi\right). \end{aligned} \tag{2.74}$$

1. A minor mismatch remains in the real part of the period when evaluated at $(m, u) = (1, u_*(1))$.
2. In contrast, the discrepancy in the imaginary part noted in [5] for $(m, u) = (2, u_*(2))$ is fully resolved by the above formula.
3. The expressions for α and β in terms of b given in eq. (2.70) apply only to the branch corresponding to $m = 1$ with $u = u_*(1) \approx 0.26318$.

2.4.1 Analysis in the (u, ℓ) coordinates

The mirror curve can be written as follows

$$\frac{\ell(y+1)}{xy} - \frac{1}{\mathrm{u}} + x + y = 0, \tag{2.75}$$

The following birational change of coordinates

$$x = \frac{bY - \frac{a}{2}}{c + X} + \frac{1}{2\mathrm{u}}, \quad y = \frac{a}{c+X}, \tag{2.76}$$

with

$$a = -36\ell \mathrm{u}^2, \quad b = \frac{1}{12\mathrm{u}}, \quad c = 12\ell\mathrm{u}^2 - 3, \tag{2.77}$$

can be used to convert eq. (2.75) to short Weierstrass form

$$Y^2 = 4X^3 - g_2(\ell, \mathrm{u})X - g_3(\ell, \mathrm{u}), \tag{2.78}$$

with

$$\begin{aligned} g_2 &= 108\left(16\ell^2\mathrm{u}^4 - 8\ell(3\mathrm{u}+1)\mathrm{u}^2 + 1\right), \\ g_3 &= -216\left(-64\ell^3\mathrm{u}^6 + 24\ell^2\left(9\mathrm{u}^2 + 6\mathrm{u} + 2\right)\mathrm{u}^4 - 12\ell(3\mathrm{u}+1)\mathrm{u}^2 + 1\right). \end{aligned} \tag{2.79}$$

As a result, we find

$$\Delta(\ell, \mathrm{u}) = g_2^3 - 27g_3^2 = \#\ell^3\mathrm{u}^8\,\delta(\ell, \mathrm{u}), \tag{2.80}$$

$$\delta(\ell, \mathrm{u}) = \ell(16\ell - 27)\mathrm{u}^4 - 36\ell\mathrm{u}^3 - 8\ell\mathrm{u}^2 + \mathrm{u} + 1, \tag{2.81}$$

$$J(\ell, \mathrm{u}) = 1728\frac{g_2^3}{\Delta} = \frac{\left(16\ell^2\mathrm{u}^4 - 8\ell(3\mathrm{u}+1)\mathrm{u}^2 + 1\right)^3}{\ell^3\mathrm{u}^8\left(\ell(16\ell-27)\mathrm{u}^4 - 36\ell\mathrm{u}^3 - 8\ell\mathrm{u}^2 + \mathrm{u} + 1\right)} \tag{2.82}$$

which matches eq. (2.59) and eq. (2.60) upto scaling.

Using the same prescription as before, we can pararmetrize the nodal curve (on the discriminant locus) as

$$x = \frac{36\ell\mathrm{u}^2(3\mathrm{u}+1) + t^3 + 3t^2 - 3tX_0 - 6X_0 - 9}{6\mathrm{u}\left(12\ell\mathrm{u}^2 + t^2 - 2X_0 - 3\right)}, \quad y = -\frac{36\ell\mathrm{u}^2}{12\ell\mathrm{u}^2 + t^2 - 2X_0 - 3} \tag{2.83}$$

with

$$X_0 = -\frac{3}{2}\frac{g_3}{g_2} = \frac{3\left(-64\ell^3\mathrm{u}^6 + 24\ell^2\left(9\mathrm{u}^2 + 6\mathrm{u} + 2\right)\mathrm{u}^4 - 12\ell(3\mathrm{u}+1)\mathrm{u}^2 + 1\right)}{16\ell^2\mathrm{u}^4 - 8\ell(3\mathrm{u}+1)\mathrm{u}^2 + 1} \tag{2.84}$$

We define $t_1 = \sqrt{-12\ell\mathrm{u}^2 + 2X_0 + 3}$. On the locus $\delta(\ell, \mathrm{u}) = 0$, it is possible to write

$$\ell = \frac{27(t_1+1)^3(t_1+3)}{16t_1^4}, \quad \mathrm{u} = \frac{2t_1^2}{9(t_1+1)}, \quad \text{on} \quad \delta(\ell, \mathrm{u}) = 0 \tag{2.85}$$

Thus, we can write the nodal curve eq. (2.83) as

$$x = \frac{(t-t_1)(t+2t_1+3)}{6\mathrm{u}(t+t_1)}, \quad y = -\frac{36\ell\mathrm{u}^2}{t^2-t_1^2} \tag{2.86}$$

In terms of t_1, X_0 takes the simple form

$$X_0 = t_1(t_1+2) \tag{2.87}$$

In order to apply the Kerr-Doran prescription, we change to a parametrization z such that the node corresponds to $z=0$ and $z=\infty$,

$$z = \frac{t-\sqrt{3X_0}}{t+\sqrt{3X_0}}. \tag{2.88}$$

The result is as follows

$$x = A\frac{(z-\frac{1}{b^2})(z-b)}{(z-1)(z-\frac{1}{b})}, \quad y = B\frac{(z-1)^2}{(z-b)(z-\frac{1}{b})}, \tag{2.89}$$

with

$$A = -\frac{b\left(b^2+b+1\right)}{(b+1)^4}, \quad B = -\frac{b^2+b+1}{(b+1)^2} \tag{2.90}$$

where we define

$$b = z(t_1) = \frac{t_1-\sqrt{3X_0}}{t_1+\sqrt{3X_0}}, \quad t_1 = -\frac{3(b+1)^2}{b^2+4b+1} \tag{2.91}$$

Note that there is a symmetry $b \to \frac{1}{b}$ due to the square root in the definition of $z(t_1)$. We can thus restrict to $|b|>1$. In terms of b, eq. (2.85) takes the form

$$\ell = -\frac{b\left(b^2+b+1\right)^3}{(b+1)^8}, \quad \mathrm{u} = -\frac{(b+1)^4}{\left(b^2+b+1\right)\left(b^2+4b+1\right)}, \quad \text{on} \quad \delta(\ell,\mathrm{u}) = 0. \tag{2.92}$$

For a fixed value of ℓ, the first equation gives eight roots in total which are invariant under $b \to \frac{1}{b}$. Thus, only four of them satisfy the condition $|b|>1$ and correspond to the four I_1 fibers for generic values of ℓ. Note that the Argeres-Douglas point $\ell = -\frac{27}{256}$, $\mathrm{u} = -\frac{8}{9}$ corresponds precisely to $b=1$. Whereas, the local $\mathbb{P}^2$ limit seems to be $b \to e^{\frac{2\pi\mathrm{i}}{3}}$.

Applying the Kerr-Doran formula, we obtain

$$Q(b) = \tag{2.93}$$

2.4.2 Differential equations satisfied by Kerr–Doran Q

Starting from the Picard–Fuchs system in the (ℓ, u) coordinates given in eq. (2.170), we change variables to the Kerr–Doran parameter b using eq. (2.92). We also introduce a small displacement variable $\delta\mathrm{u}$ transverse to the discriminant locus, so that

$$\ell = \ell(b), \qquad \mathrm{u} = \mathrm{u}(b) + \delta\mathrm{u}.$$

Rewriting the Picard–Fuchs operators in the coordinates $(b, \delta\mathrm{u})$ and expanding near the discriminant to leading order in $\delta\mathrm{u}$, we expand a local period in the form

$$G(b, \delta\mathrm{u}) = G_0(b) + G_1(b)\,\delta\mathrm{u} + O(\delta\mathrm{u}^2).$$

At leading order, the transformed Picard–Fuchs equations involve $G_1(b)$ and $G_1'(b)$. However, by taking an appropriate linear combination of the two equations, these auxiliary terms can be eliminated, yielding a single second-order ODE for the cusp period $G_0(b)$:

$$G_0''(b) + \frac{b^4 - 4b^3 - 8b^2 - 6b - 1}{b\,(b^4 + b^3 - b - 1)}\,G_0'(b) = 0. \tag{2.94}$$

2.4.3 Domain, range, and branching structure of the map $\ell(b)$

Consider the rational map of Riemann spheres

$$\ell : \mathbb{P}^1_b \longrightarrow \mathbb{P}^1_\ell, \qquad \ell(b) \;=\; -\frac{b\,(1 + b + b^2)^3}{(1 + b)^8}\,. \tag{2.95}$$

Since ℓ is a non-constant holomorphic map $\mathbb{P}^1 \to \mathbb{P}^1$, it is surjective onto $\mathbb{P}^1_\ell$; the interesting features are its divisor, critical points, and branch values.

Divisor data: zeros and poles. The only pole occurs at $b = -1$, with order 8:

$$\ell(b) \sim -\frac{(-1)\,(1 - 1 + 1)^3}{(b + 1)^8} \quad \text{as } b \to -1, \qquad \Rightarrow \qquad \mathrm{ord}_{b=-1}(\ell) = -8, \;\; \ell(-1) = \infty. \tag{2.96}$$

The numerator vanishes at $b = 0$ (simple) and at the primitive cube roots of unity $b = \omega, \omega^2$ (each with multiplicity 3). In addition, as $b \to \infty$,

$$\ell(b) = -\frac{b(b^2)^3}{b^8}\big(1 + O(1/b)\big) = -\frac{1}{b}\big(1 + O(1/b)\big), \tag{2.97}$$

so $\ell(\infty) = 0$ and the zero at ∞ is simple. Thus

$$\ell^{-1}(0) = \{0, \infty, \omega, \omega^2\} \quad \text{with multiplicities} \quad 1, 1, 3, 3, \tag{2.98}$$

and the total multiplicity equals 8, anticipating $\deg(\ell) = 8$.

Degree. The map has a unique pole of order 8 at $b = -1$, hence

$$\deg(\ell) = 8. \tag{2.99}$$

Equivalently, for generic $\ell_0 \in \mathbb{C} \subset \mathbb{P}^1_\ell$, the equation $\ell(b) = \ell_0$ has 8 solutions in $\mathbb{P}^1_b$ counted with multiplicity.

Involution symmetry and factorization. A key simplification is the involutive symmetry

$$\ell\left(\frac{1}{b}\right) = \ell(b), \tag{2.100}$$

so ℓ factors through the $\mathbb{Z}_2$-quotient $\mathbb{P}^1_b/\langle b \mapsto 1/b\rangle \cong \mathbb{P}^1_u$ via the standard invariant coordinate

$$u \;:=\; b + \frac{1}{b}. \tag{2.101}$$

Using the identities $b^2 + b + 1 = b(u + 1)$ and $(b + 1)^8 = b^4(u + 2)^4$, one obtains the compact expression

$$\ell(b) = -\frac{(u + 1)^3}{(u + 2)^4}, \qquad u = b + \frac{1}{b}. \tag{2.102}$$

Hence the map decomposes as

$$\mathbb{P}^1_b \xrightarrow{\deg 2} \mathbb{P}^1_u \xrightarrow{\deg 4} \mathbb{P}^1_\ell, \tag{2.103}$$

and the total degree is $2 \cdot 4 = 8$.

Critical points and ramification. Differentiating, one finds

$$\frac{d\ell}{db} = \frac{(b - 1)^3\,(b^2 + b + 1)^2}{(b + 1)^9}. \tag{2.104}$$

Thus the critical points in $\mathbb{P}^1_b$ are

$$b = 1, \qquad b = \omega, \omega^2, \qquad b = -1, \tag{2.105}$$

where $b = -1$ is the pole (and hence a branch point in the sense of the map to $\mathbb{P}^1_\ell$). From the local behavior implied by (2.104) (or directly by series expansion), the ramification indices are:

$$e_{-1} = 8, \qquad \ell(-1) = \infty \quad \text{(totally ramified)}, \tag{2.106}$$

$$e_1 = 4, \qquad \ell(1) = -\frac{27}{256}, \tag{2.107}$$

$$e_\omega = e_{\omega^2} = 3, \qquad \ell(\omega) = \ell(\omega^2) = 0. \tag{2.108}$$

In particular, ℓ has exactly three branch values in $\mathbb{P}^1_\ell$:

$$\mathrm{Branch}(\ell) = \left\{0,\ -\frac{27}{256},\ \infty\right\}. \tag{2.109}$$

Rescaling $\lambda := -(256/27)\,\ell$ gives a Belyi normalization with branch locus $\{0, 1, \infty\}$.

Fibers over the branch values. The full fiber structure (with multiplicities) over the three branch values is:

$$\ell^{-1}(\infty) = \{-1\} \quad \text{with multiplicity } 8, \tag{2.110}$$

$$\ell^{-1}(0) = \{0, \infty, \omega, \omega^2\} \quad \text{with multiplicities } 1, 1, 3, 3, \tag{2.111}$$

$$\ell^{-1}\left(-\frac{27}{256}\right) = \{1\} \ \cup\ \{\text{four further points}\}, \tag{2.112}$$

where $b = 1$ appears with multiplicity 4 (ramification index 4), and the remaining four simple preimages are the roots of the reciprocal quartic

$$27b^4 + 68b^3 + 98b^2 + 68b + 27 = 0. \tag{2.113}$$

Riemann–Hurwitz check. For a degree-8 map $\mathbb{P}^1 \to \mathbb{P}^1$, Riemann–Hurwitz gives

$$\sum_{p \in \mathbb{P}^1_b} \big(e_p - 1\big) = 2\deg(\ell) - 2 = 14. \tag{2.114}$$

Using the above ramification indices,

$$(8-1) + (4-1) + (3-1) + (3-1) = 7 + 3 + 2 + 2 = 14, \tag{2.115}$$

so the branching data is consistent.

Real locus: range on $\mathbb{P}^1(\mathbb{R})$. Restricting to $b \in \mathbb{R} \cup \{\infty\}$, the invariant coordinate satisfies $u = b + 1/b \in (-\infty, -2] \cup [2, \infty)$ (for $b \neq 0$), and (2.102) shows $\ell(b) \in \mathbb{R}$. More precisely:

- For $u \in [2, \infty)$ (equivalently $b > 0$), ℓ increases from $\ell(1) = -27/256$ to $\ell(\infty) = 0$, hence

$$\ell\big(\mathbb{R}_{>0} \cup \{\infty\}\big) = [-27/256,\ 0].$$

- For $u \in (-\infty, -2)$ (equivalently $b < 0$, $b \neq -1$), ℓ decreases from $+\infty$ to 0^+, with the pole at $b = -1$, hence

$$\ell\big(\mathbb{R}_{<0}\big) = (0, \infty), \qquad \ell(-1) = \infty.$$

In particular, no real b maps to $\ell < -27/256$; crossing the real threshold $\ell = -27/256$ requires complex b.

Summary. The map $\ell(b)$ is a degree-8 rational function on $\mathbb{P}^1_b$, invariant under $b \mapsto 1/b$, factoring through $u = b + 1/b$ as

$$\ell = -\frac{(u+1)^3}{(u+2)^4}.$$

Its branch locus in $\mathbb{P}^1_\ell$ is precisely $\{0, -27/256, \infty\}$ with ramification profile

$$\infty:\ 8; \qquad -\frac{27}{256}:\ 4, 1, 1, 1, 1; \qquad 0:\ 3, 3, 1, 1,$$

and this data satisfies Riemann–Hurwitz.

2.5 Large Volume Central Charge

We take $\mathbb{F}_1$ to be the blow-up of $\mathbb{P}^2$. We define a basis of compact curve classes on local $\mathbb{F}_1$. Let C be the exceptional class, $C^2 = -1$ in the cup product. Let H be the hyperplane class, $H^2 = 1$ and $H.C = 0$. We consider compactly supported coherent sheaves E on local $\mathbb{F}_1$ with Chern vector $\gamma(E) = [r(E),\ c_1(E),\ \mathrm{ch}_2(E)]$, where r is the rank of the sheaf on $\mathbb{F}_1$, $c_1(E)$ is the first Chern class

2.6 To organize

Sending $\log(\tilde{u}) \to \log(-\tilde{u})$ in the solutions ϖ_2 and ϖ_3 of the PF system transforms (T, T_D) as follows

$$T \to T - \frac{1}{2}, \quad T_D \to T_D - 4T - \frac{m}{2} + 1. \tag{2.116}$$

which at the level of charges corresponds to

$$(r, d_F, d_B, \mathrm{ch}_2) \;\to\; \big(r,\ d_F + \tfrac{3}{2}r,\ d_B + r,\ \mathrm{ch}_2 + d_F + \tfrac{1}{2}d_B + r\big) \tag{2.117}$$

Whereas, under the large volume monodromies sending $\log(\tilde{u}) \to \log(\tilde{u}) + 2\pi \mathrm{i} k$, we get

$$T \to T - k, \quad T_D \to T_D - 8kT - km + 4k^2. \tag{2.118}$$

and the coordinates (x, y) transform to

$$x \to x - k, \quad y \to y + (8x + \mu)k - 4k^2 \tag{2.119}$$

2.7 Large Volume Scattering

We take for the large volume central charge of an object E of chern Character $\gamma = (r,\ d_F,\ d_B,\ \mathrm{ch}_2)$, the function

$$Z_{\mathrm{LV}}(\gamma) = -r\left(4T^2 + mT - \frac{m^2}{2}\right) + d_F(2T + m) + d_B(T - m) - \mathrm{ch}_2. \tag{2.120}$$

2.7.1 Walls of Marginal Stability

Walls of marginal stability between two objects of charge γ and γ' occur when their central charges are aligned, $\arg Z_{\mathrm{LV}}(\gamma) = \arg Z_{\mathrm{LV}}(\gamma') \pmod{2\pi}$. It is possible to show that for fixed real m, walls of marginal stability are circles in the T-plane. The center is at

$$T_0 = \frac{\mathrm{ch}_2' r - \mathrm{ch}_2 r' - d_B m r' + d_B' m r - d_F' m r + d_F m r'}{\langle \gamma, \gamma' \rangle} \in \mathbb{R} \tag{2.121}$$

on the real axis, while the radius R can be written as

$$R^2 = T_0^2 + \frac{T_0(-d'_B - 2d'_F + mr')}{4r'} + \frac{2\mathrm{ch}'_2 - m(-2d'_B + 2d'_F + mr')}{8r'} \tag{2.122}$$

2.7.2 Rays

Given a phase $\psi \in [0, \pi)$, the ray corresponding to an object E of charge γ is given by

$$\text{Ray equation:} \quad \mathcal{F}(\gamma, T, m, \psi) = \mathrm{Re}(e^{-\mathrm{i}\psi} Z_{\mathrm{LV}}(\gamma)) = 0. \tag{2.123}$$

and $\mathrm{Im}(e^{-\mathrm{i}\psi} Z_{\mathrm{LV}}(\gamma)) > 0$. Of course, this ray may or may not be populated over the entire T plane but the equation above is the support of the ray whenever it exists (also for the object of charge $-\gamma$). A direct evaluation of $\mathcal{F}$ for the central charge in eq. (2.120) shows that the rays are rectangular hyperbolas in the plane of $T = s + \mathrm{i}t$. The center of the hyperbola is at

$$T_0 = \frac{1}{8}\left[\frac{(d_B + 2d_F)}{r} - m\right], \tag{2.124}$$

After performing a rotation by $\frac{\psi}{2}$ and shifting the center, defining $T' = e^{-\mathrm{i}\frac{\psi}{2}}(T - T_0) = s' + \mathrm{i}t'$ and $m = m_1 + \mathrm{i}m_2$, the ray equation looks like

$$\begin{aligned} s'^2 - t'^2 = &\frac{-2r(8\mathrm{ch}_2 + (9d_B - 6d_F)m_1) + (d_B + 2d_F)^2 + 9r^2(m_1^2 - m_2^2)}{64r^2} \cos\psi \\ &+ \frac{3m_2(-3d_B + 2d_F + 3m_1 r)}{32r} \sin\psi. \end{aligned} \tag{2.125}$$

2.7.3 Rays in x-y coordinates

We define coordinates

$$x = \frac{\mathrm{Re}(e^{-\mathrm{i}\psi}\, T)}{\cos\psi}, \quad y = -\frac{\mathrm{Re}(e^{-\mathrm{i}\psi}\, T_D)}{\cos\psi}, \quad \mu = \frac{\mathrm{Re}(e^{-\mathrm{i}\psi}\, m)}{\cos\psi}, \tag{2.126}$$

which makes the rays into straight lines

$$x(d_B + 2d_F) + ry - \mathrm{ch}_2 + \mu(d_F - d_B) = 0 \tag{2.127}$$

2.7.4 Determining initial rays from domains of validity of strongly exceptional collections

We consider three strongly exceptional collections and, for each, record the corresponding dual collection:

Bridgeland–Stern–Perling collection

$$C_B = \{\ \mathcal{O}(0,0),\ \mathcal{O}(-1,0)[1],\ \mathcal{O}(0,-1)[1],\ \mathcal{O}(-1,-1)\ \} \tag{2.128}$$

Arcara–Miles I collection

$$C_{AI} = \left\{\, \mathcal{O}(0,0),\ \mathrm{GV}(0,1,-\tfrac{1}{2}),\ \mathcal{O}(-2,-1),\ \mathcal{O}(-1,0)[1] \,\right\} \tag{2.129}$$

Arcara–Miles II collection

$$C_{AII} = \left\{\, \mathcal{O}(0,0),\ \mathrm{GV}(0,-1,\tfrac{1}{2}),\ \mathcal{O}(-2,-1),\ \mathcal{O}(-1,-1)[1] \,\right\} \tag{2.130}$$

Given a collection $C = \{\gamma_1, \dots, \gamma_n\}$, its domain of validity at phase ψ is

$$\mathrm{dom}_\psi C := \left\{\mathrm{Re}\big(e^{-\mathrm{i}\psi} Z_{\gamma_k}\big) < 0,\ \forall\, \gamma_k \in C\right\}. \tag{2.131}$$

In the x–y coordinates of eq. (2.126), and assuming $0 \le \psi < \frac{\pi}{2}$, the condition $\mathrm{Re}(e^{-\mathrm{i}\psi} Z_\gamma) < 0$ takes the form

$$r\, y + (d_B + 2d_F)\, x + (d_F - d_B)\, \mu - \mathrm{ch}_2 < 0. \tag{2.132}$$

Hence $\mathrm{dom}_\psi C$ is cut out by a finite system of linear inequalities; when non-empty and bounded it is a polygon. [Is it always bounded?]

We now study the domains of validity of the above collections after tensoring by a line bundle $\mathcal{O}(m_F, m_B)$. Equivalently, we seek the integer pairs $(m_F, m_B) \in \mathbb{Z}^2$ for which $\mathrm{dom}_0\big(C \otimes \mathcal{O}(m_F, m_B)\big) \neq \varnothing$.

Case $3\mu \notin \mathbb{Z}$. For each fixed $m_F \in \mathbb{Z}$, there is exactly one integer m_B for which the domain is non-empty:

$$C_B: \qquad m_B = \left\lceil \frac{2m_F + 3\mu - 1}{3} \right\rceil, \tag{2.133}$$

$$C_{AI}: \qquad m_B = \left\lceil \frac{2m_F + 3\mu - 3}{3} \right\rceil, \tag{2.134}$$

$$C_{AII}: \qquad m_B = \left\lceil \frac{2m_F + 3\mu - 1}{3} \right\rceil. \tag{2.135}$$

Case $3\mu \in \mathbb{Z}$. In this case, there is no spectral flow in (m_F, m_B) whenever the argument of the corresponding ceiling is already an integer (i.e. whenever the ceiling acts trivially). Concretely, for the above formulas this means:

- for C_B and C_{AII}, no spectral flow occurs whenever $\frac{2m_F+3\mu-1}{3} \in \mathbb{Z}$;
- for C_{AI}, no spectral flow occurs whenever $\frac{2m_F+3\mu-3}{3} \in \mathbb{Z}$.

Charge (FB)	Charge (HC)	Contributing tree(s)	Index $\Omega_H(y)$ (nonneg. part)
$[0,0,1,\frac{1}{2}]$	$[0,0,1,\frac{1}{2}]$	$\{\mathcal{O}(1,0)[1],\,\mathcal{O}(1,1)\}$	1
$[0,1,0,0]$	$[0,1,-1,0]$	$\{\mathcal{O}(0,0)[1],\,\mathcal{O}(1,0)\}$	$-y$
$[0,1,1,\frac{1}{2}]$	$[0,1,0,\frac{1}{2}]$	$\{\mathcal{O}(0,0)[1],\,\mathcal{O}(1,1)\}$	y^2+1
$[0,2,1,\frac{1}{2}]$	$[0,2,-1,\frac{1}{2}]$	$\{\mathcal{O}(1,1),\,\{2\,\mathcal{O}(0,0)[1],\,\mathcal{O}(1,0)\}\}$	y^4+y^2+1
$[0,2,2,0]$	$[0,2,0,0]$	$\{\mathcal{O}(1,1),\,\mathcal{O}(-1,-1)[1]\}$	$-y^5-y^3-y$
$[0,3,2,0]$	$[0,3,-1,0]$	$\{\mathcal{O}(-1,-1)[1],\,\{\mathcal{O}(1,1),\,\{\mathcal{O}(0,0)[1],\,\mathcal{O}(1,0)\}\}\}$	$-y^9-3y^7-4y^5-4y^3-4y$
$[0,3,3,\frac{1}{2}]$	$[0,3,0,\frac{1}{2}]$	$\{\mathcal{O}(-1,-1)[1],\,\{\mathcal{O}(0,0)[1],\,2\,\mathcal{O}(1,1)\}\}$	$y^{10}+2y^8+3y^6+3y^4+3y^2+3$
$[0,4,2,0]$	$[0,4,-2,0]$	① $\{\mathcal{O}(-2,-1)[1],\,\mathcal{O}(2,1)\}$ ② $\{\mathcal{O}(-1,-1)[1],\,\{\mathcal{O}(1,1),\,\{2\,\mathcal{O}(0,0)[1],\,2\,\mathcal{O}(1,0)\}\}\}$	$-y^{13}-3y^{11}-8y^9-10y^7-11y^5-11y^3-11y$
$[1,2,2,2]$	$[1,2,0,2]$	$\{\mathcal{O}(2,1),\,\{\mathcal{O}(1,0)[1],\,\mathcal{O}(1,1)\}\}$	1
$[1,0,1,-\frac{1}{2}]$	$[1,0,1,-\frac{1}{2}]$	$\{\mathcal{O}(0,0),\,\{\mathcal{O}(-2,-2)[1],\,\mathcal{O}(-2,-1)\}\}$	1
$[1,1,2,0]$	$[1,1,1,0]$	$\{\mathcal{O}(1,1),\,\{\mathcal{O}(-2,-2)[1],\,\mathcal{O}(-2,-1)\}\}$	1
$[1,2,3,\frac{3}{2}]$	$[1,2,1,\frac{3}{2}]$	$\{\{\mathcal{O}(-2,-2)[1],\,\mathcal{O}(-2,-1)\},\,\{\mathcal{O}(2,1),\,\{\mathcal{O}(1,0)[1],\,\mathcal{O}(1,1)\}\}\}$	1
$[-1,0,1,\frac{1}{2}]$	$[-1,0,1,\frac{1}{2}]$	$\{\mathcal{O}(0,0)[1],\,\{\mathcal{O}(1,0)[1],\,\mathcal{O}(1,1)\}\}$	1
$[-1,-2,-1,-\frac{3}{2}]$	$[-1,-2,1,-\frac{3}{2}]$	$\{\mathcal{O}(2,1)[1]\}$	1
$[1,0,0,-1]$	$[1,0,0,-1]$	① $\{\mathcal{O}(-2,-2)[1],\,2\,\mathcal{O}(-1,-1)\}$ ② $\{\{\mathcal{O}(-2,-2)[1],\,\mathcal{O}(-2,-1)\},\,\{\mathcal{O}(-2,-1)[1],\,2\,\mathcal{O}(-1,-1)\}\}$	y^2+2
$[1,0,0,-2]$	$[1,0,0,-2]$	① $\{\mathcal{O}(-1,-1),\,\{\mathcal{O}(-3,-2)[1],\,\mathcal{O}(-2,-1)\}\}$ ② $\{\{\mathcal{O}(-5,-4)[1],\,\mathcal{O}(-5,-3)\},\,\{\mathcal{O}(-2,-1)[1],\,2\,\mathcal{O}(-1,-1)\}\}$ ③ $\{\{\mathcal{O}(-3,-2)[1],\,\mathcal{O}(-2,-2)\},\,\{\mathcal{O}(-1,-1),\,\{\mathcal{O}(-2,-2)[1],\,\mathcal{O}(-2,-1)\}\}\}$	y^4+3y^2+6
$[2,1,1,-\frac{1}{2}]$	$[2,1,0,-\frac{1}{2}]$	$\{3\,\mathcal{O}(0,0),\,\mathcal{O}(-1,-1)[1]\}$	1
$[2,1,0,-1]$	$[2,1,-1,-1]$	$\{2\,\mathcal{O}(0,0),\,\{\mathcal{O}(-2,-1)[1],\,\mathcal{O}(-1,-1)\}\}$	$-y$

Table 1: Charges, contributing trees, and the nonnegative-power part of the refined index $\Omega_H(y)$ (the full Laurent polynomial is symmetric under $y \leftrightarrow y^{-1}$) for $m = -\frac{1}{10}$.

2.8 Cusps in the U-plane

The discriminant of the mirror curve

$$\mathrm{Disc}(\mathbb{F}_1) = u^8\Big((16\lambda^3-27)u^4 - 36\lambda u^3 - 8\lambda^2u^2 + u + \lambda\Big) \tag{2.136}$$

has five roots, the large volume point (or point of maximal unipotent monodromy, MUM) at $u=0$ (which is an I_8 fiber) and four standard conifold points which we denote $u_1,\ldots,u_4$. These are I_1 fibers for generic λ. At $\lambda = 1$, these points are plotted in fig. 3. We generally label the conifold points in this order for $\lambda = 1$ and by continuation to any other λ avoiding the singular locus ($\lambda^3 = -27/256$).

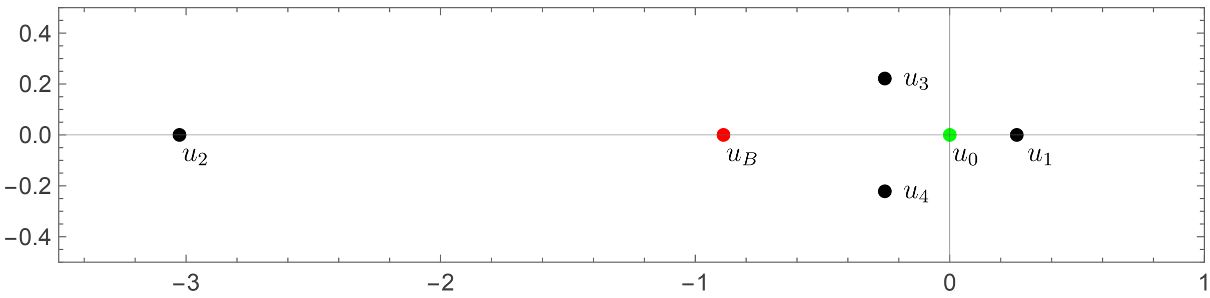


Figure 3: Singularities in the U-plane and the branch point.

To see that that these are also the singularities of the Picard-Fuchs equations (and additional singularities), we write the third-order PF equations (for periods of the CY3) in this form

$$\partial_u^3 \varpi(\lambda, u) \; + \; P(\lambda, u)\, \partial_u^2 \varpi(\lambda, u) \; + \; Q(\lambda, u)\, \partial_u \varpi(\lambda, u) = 0 \tag{2.137}$$

where P and Q are the following rational functions

$$P = \frac{24\lambda^2 + 54\left(16\lambda^3 - 27\right) u^5 + 4\lambda\left(224\lambda^3 - 783\right) u^4 - 2016\lambda^2 u^3 + \left(27 - 320\lambda^3\right) u^2 + 50\lambda u}{u(8\lambda + 9u)\left(\lambda + \left(16\lambda^3 - 27\right) u^4 - 36\lambda u^3 - 8\lambda^2 u^2 + u\right)} \tag{2.138}$$

$$Q = \frac{8\lambda^2 + 54\left(16\lambda^3 - 27\right) u^5 + 16\lambda\left(64\lambda^3 - 189\right) u^4 - 1860\lambda^2 u^3 + \left(9 - 256\lambda^3\right) u^2 + 16\lambda u}{u^2(8\lambda + 9u)\left(\lambda + \left(16\lambda^3 - 27\right) u^4 - 36\lambda u^3 - 8\lambda^2 u^2 + u\right)} \tag{2.139}$$

From this, it is clear that the only singular points of the third-order PF are $u_0 = 0,\ u_1,\ u_2,\ u_3,\ u_4,$ and $u_B = -8\lambda/9$. It is also clear that these are all regular singular points. At $u = 0$, the indicial polynomial is $r^3 = 0$, consistent with it being a MUM point. At $u = u_B = -8\lambda/9$, the indicial polynomial turns out to be $(-3 + r)(-1 + r)r$. At $u = u_i$, $i = 1, 2, 3, 4$, the indicial polynomial is $(-1 + r)^2 r$. At $u = 0$, if we pick a local Frobenius basis

$$\varpi_0 = 1, \quad \varpi_1 = \log\lambda = 2\pi \mathrm{i} m, \tag{2.140}$$

$$\varpi_2 = \log(u) + (\text{analytic}), \tag{2.141}$$

$$\varpi_3 = (\log(u))^2 + \varpi_2(-2\log u - \frac{1}{4}\log\lambda) + \frac{(\log\lambda)^2}{8} + (\text{analytic}), \tag{2.142}$$

then the large volume periods T and T_D can be written as

$$T = \frac{\varpi_2}{(2\pi \mathrm{i})}, \quad T_D = -4\frac{\varpi_3}{(2\pi \mathrm{i})^2} + \frac{1}{6}. \tag{2.143}$$

Method for computing monodromies

To numerically compute monodromies around the singular points, I define the companion system to eq. (2.137),

$$\partial_u Y(u) = A(u) Y(u) \tag{2.144}$$

where we define the system vector $Y(u)$ and the connection matrix $A(u)$ as,

$$Y(u) = \begin{pmatrix} \varpi(u) \\ \partial_u \varpi(u) \\ \partial_u^2 \varpi(u) \end{pmatrix}, \quad A(u) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -P(u) & -Q(u) \end{pmatrix}. \tag{2.145}$$

We define the fundamental matrix $\Phi(u)$ for the fundamental solutions $\varpi_1(u), \varpi_2(u), \varpi_3(u)$ as

$$\Phi(u) = \begin{pmatrix} \varpi_1 & \varpi_2 & \varpi_3 \\ \partial_u \varpi_1 & \partial_u \varpi_2 & \partial_u \varpi_3 \\ \partial_u^2 \varpi_1 & \partial_u^2 \varpi_2 & \partial_u^2 \varpi_3 \end{pmatrix}, \tag{2.146}$$

which also satisfies $\partial_u \Phi(u) = A(u)\Phi(u)$. Thus, to compute monodromies around closed paths $\gamma : [0,1] \to \mathbb{P}^1/\{u_i\}, t \mapsto u(t)$, we can numerically solve the ODE

$$\dot{\Psi}(t) = \dot{u}(t)\, A(u(t))\Psi(t), \tag{2.147}$$

with any initial condition e.g. ($\Psi(0) = I_3$), and then compute the change-of-basis matrix C at the base-point $u_* = u(0)$ which we can take to be inside a small disc around $u_0 = 0$ where we know the period expansions well. The matrix C is given by

$$C = \begin{pmatrix} 1 & T(u_*) & T_D(u_*) \\ 0 & \partial_u T(u_*) & \partial_u T_D(u_*) \\ 0 & \partial_u^2 T(u_*) & \partial_u^2 T_D(u_*) \end{pmatrix}, \tag{2.148}$$

and the local monodromy around path γ is given by

$$M_\gamma = C^{-1}\Psi(1)C. \tag{2.149}$$

Monodromies at $\lambda = 1$

Using the procedure described above, we find the following monodromies in the $(1, m, T, T_D)$ basis where the second column is deduced by changing m slightly and comparing the first column of the 3×3 monodromy on $(1, T, T_D)$ computed by numerics

Around $u = 0$:

$$M_{LV} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 4 & 8 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 4 & 1 & 8 & 1 \end{pmatrix} \tag{2.150}$$

Around $u = u_B = -8\lambda/9$:

$$M_B = I_3 \;\to\; I_4 \tag{2.151}$$

2.9 Degenerations of the map $\mathrm{u} \mapsto J$

Fix the parameter $\ell \in \mathbb{C}$ and consider the rational map

$$J(\mathrm{u};\ell) = \frac{N(\mathrm{u};\ell)^3}{\ell^3\,\mathrm{u}^8\,E(\mathrm{u};\ell)}, \tag{2.152}$$

where

$$N(\mathrm{u};\ell) := 1 + 16\ell^2\mathrm{u}^4 - 8\ell\mathrm{u}^2(1+3\mathrm{u}), \tag{2.153}$$
$$E(\mathrm{u};\ell) := 1 + \mathrm{u} - 8\ell\mathrm{u}^2 - 36\ell\mathrm{u}^3 + \ell(-27+16\ell)\mathrm{u}^4. \tag{2.154}$$

For generic ℓ, the map (2.152) defines a degree-12 cover $\phi_\ell : \mathbb{P}^1_{\mathrm{u}} \to \mathbb{P}^1_J$ (i.e. generically 12 preimages for a given J).

2.9.1 Branch values via the discriminant method

For fixed ℓ, branch values J_* are characterized by the condition that the fiber $\phi_\ell^{-1}(J_*)$ has multiplicities, i.e. that the equation $J(\mathrm{u};\ell) = J$ has a multiple root in u. Equivalently, define the polynomial in u

$$P(\mathrm{u};J,\ell) := N(\mathrm{u};\ell)^3 - J\,\ell^3\mathrm{u}^8E(\mathrm{u};\ell). \tag{2.155}$$

Then J is a branch value of ϕ_ℓ if and only if

$$\mathrm{Disc}_{\mathrm{u}}\big(P(\mathrm{u};J,\ell)\big) = 0. \tag{2.156}$$

After factoring, one finds that (up to an overall nonzero constant)

$$\mathrm{Disc}_{\mathrm{u}}\big(P(\mathrm{u};J,\ell)\big) = J^8(J-1728)^6\big(J-J_B(\ell)\big)\cdot\ell^{44}(256\ell+27)^3. \tag{2.157}$$

Hence for generic $\ell \neq 0, -\frac{27}{256}$ the branch values on the J-sphere are

$$J \in \{0,\ 1728,\ J_B(\ell),\ \infty\}, \qquad J_B(\ell) = \frac{(256\ell+27)(256\ell+243)^3}{2^{24}\,\ell^3}. \tag{2.158}$$

2.9.2 Critical points and generic ramification profile

Critical points in the u-sphere are the zeros of $\frac{dJ}{d\mathrm{u}}$. Writing $\frac{dJ}{d\mathrm{u}} = \frac{\mathrm{num}}{\mathrm{den}}$, the numerator factorizes as

$$\mathrm{num}\left(\frac{dJ}{d\mathrm{u}}\right) = \ell^3\mathrm{u}^7(9\mathrm{u}+8)\,Q(\mathrm{u};\ell)^2\,S(\mathrm{u};\ell), \tag{2.159}$$

where

$$Q(\mathrm{u};\ell) := 16\ell^2\mathrm{u}^4 - 24\ell\mathrm{u}^3 - 8\ell\mathrm{u}^2 + 1, \tag{2.160}$$
$$S(\mathrm{u};\ell) := 64\ell^3\mathrm{u}^6 - 216\ell^2\mathrm{u}^6 - 144\ell^2\mathrm{u}^5 - 48\ell^2\mathrm{u}^4 + 36\ell\mathrm{u}^3 + 12\ell\mathrm{u}^2 - 1. \tag{2.161}$$

For generic ℓ, the ramification data over the branch values in (2.158) is:

- Over $J = \infty$: $\mathrm{u} = 0$ is an order-8 pole, hence ramification index $e = 8$ (contribution $e - 1 = 7$). The remaining poles are the simple roots of $E(\mathrm{u}; \ell)$.
- Over $J = 0$: $N(\mathrm{u}; \ell) = 0$ is equivalent to $Q(\mathrm{u}; \ell) = 0$ (since N is quartic), and the square Q^2 in (2.159) implies that each root of Q is ramified with $e = 3$. Thus the fiber over 0 has cycle type
$$3 + 3 + 3 + 3.$$
- Over $J = 1728$: the six roots of $S(\mathrm{u}; \ell)$ map to 1728 and are simple critical points, giving cycle type
$$2 + 2 + 2 + 2 + 2 + 2.$$
- Over $J = J_B(\ell)$: the factor $(9\mathrm{u} + 8)$ shows that $\mathrm{u} = -\frac{8}{9}$ is a simple critical point, hence contributes one transposition; the remaining ten points are unramified. Cycle type:
$$2 + 1 + \cdots + 1 \quad \text{(ten 1's)}.$$

2.9.3 What it means for the map to "degenerate"

There are two distinct sources of degeneration.

(I) Special branch values for fixed ℓ. For fixed ℓ, the fiber over J ceases to consist of 12 distinct points precisely when (2.156) holds. The branch values are therefore (2.158), and multiplicities are encoded in the factorization of the discriminant (2.157).

(II) Special parameters ℓ where the cover changes type. As ℓ varies, the Hurwitz data (cycle types, collisions of branch values, degree) can change at parameter values where:

1. **Cancellation (degree drop):** numerator and denominator share a common factor, i.e. $\gcd\big(N(\mathrm{u}; \ell)^3, \ell^3 \mathrm{u}^8 E(\mathrm{u}; \ell)\big) \neq 1$. Equivalently, $\mathrm{Res}_{\mathrm{u}}\big(N(\mathrm{u}; \ell), u^8 E(\mathrm{u}; \ell)\big) = 0$.
2. **Critical points collide:** $\mathrm{Disc}_{\mathrm{u}}(Q(\mathrm{u}; \ell)) = 0$ or $\mathrm{Disc}_{\mathrm{u}}(S(\mathrm{u}; \ell)) = 0$.
3. **Branch values collide:** $J_B(\ell)$ hits $\{0, 1728, \infty\}$ or merges with the value $J(\infty)$ (see below).

2.9.4 Distinguished special values of ℓ (explicit list)

The following parameter values are singled out by the above mechanisms:

- **Ill-defined family point:** $\ell = 0$ (the formula (2.152) has ℓ^3 in the denominator).
- **True cancellation / degree drop:**
$$\ell = -\frac{27}{256}. \tag{2.162}$$
At this value numerator and denominator share a nontrivial common factor in u, so after cancellation the degree drops below 12.

- **Collision in the $J = 0$ fiber (ramification increases):**
$$\ell = -\frac{243}{256}. \tag{2.163}$$
Here $Q(\mathrm{u};\ell)$ acquires a repeated root at $\mathrm{u} = -\frac{8}{9}$, changing the cycle type over $J = 0$.
- **Collision in the $J = 1728$ fiber and branch-value collision $J_B(\ell) = 1728$:**
$$\ell = \frac{243}{256} \pm \frac{81\sqrt{3}}{128}, \tag{2.164}$$
equivalently the roots of $65536\ell^2 - 124416\ell - 19683 = 0$. These are characterized by $\mathrm{Disc}_{\mathrm{u}}(S(\mathrm{u};\ell)) = 0$ and simultaneously $J_B(\ell) = 1728$.

2.9.5 Behavior at $u = \infty$ and additional special values

The value of the map at infinity is obtained from the leading terms:
$$J(\infty;\ell) = \lim_{u\to\infty} J(u;\ell) = \frac{4096\,\ell^2}{16\ell - 27}, \qquad (\ell \neq 27/16). \tag{2.165}$$
This exhibits further distinguished parameters:

- $\ell = \frac{27}{16}$: the leading term of $E(\mathrm{u};\ell)$ vanishes, so $J(\infty;\ell) = \infty$ (i.e. a pole effectively moves to $\mathrm{u} = \infty$).
- $\ell = \frac{27}{8}$: one has $J(\infty;\ell) = 1728$; in addition $\mathrm{u} = \infty$ becomes ramified (the local expansion starts at order $(1/\mathrm{u})^2$), producing extra branching over $J = 1728$.

2.9.6 Periods near $\mathrm{u} = -\frac{8}{9}$

The third order Picard-Fuchs equation for the periods of local $\mathbb{F}_1$ can be written as $\mathcal{D}_3\varpi(\ell,\mathrm{u}) = 0$, with
$$\mathcal{D}_3 = \partial_{\mathrm{u}}^3 + P(\ell,\mathrm{u})\partial_{\mathrm{u}}^2 + Q(\ell,\mathrm{u})\partial_{\mathrm{u}}, \tag{2.166}$$
$$\begin{aligned}
P(\ell,\mathrm{u}) &= \frac{54\ell(16\ell-27)\mathrm{u}^5 + 4\ell(224\ell-783)\mathrm{u}^4 - 2016\ell\mathrm{u}^3 + (27-320\ell)\mathrm{u}^2 + 50\mathrm{u} + 24}{\delta(\ell,\mathrm{u})},\\
Q(\ell,\mathrm{u}) &= \frac{54\ell(16\ell-27)\mathrm{u}^5 + 16\ell(64\ell-189)\mathrm{u}^4 - 1860\ell\mathrm{u}^3 + (9-256\ell)\mathrm{u}^2 + 16\mathrm{u} + 8}{\mathrm{u}\,\delta(\ell,\mathrm{u})},\\
\delta(\ell,\mathrm{u}) &= \mathrm{u}(9\mathrm{u}+8)\left(\ell(16\ell-27)\mathrm{u}^4 - 36\ell\mathrm{u}^3 - 8\ell\mathrm{u}^2 + \mathrm{u} + 1\right)
\end{aligned} \tag{2.167}$$

The point $\mathrm{u} = -\frac{8}{9}$ is a regular singular point with local exponents $(0,1,3)$ for generic values of ℓ and $(0,\frac{5}{6},\frac{7}{6})$ at $\ell = -\frac{27}{256}$. The solutions can be written in terms of $\mathrm{u} = -8/9 + z$ in the canonical Frobenius basis as follows

$$
\begin{aligned}
\varpi_0 &= 1, \\
\varpi_1 &= z + \frac{9(512\ell+27)z^2}{4096\ell+432} - \frac{729\left(67108864\ell^3-4423680\ell^2-559872\ell-45927\right)z^4}{1024(256\ell+27)^3} + \mathcal{O}(z^5), \\
\varpi_2 &= z^3 + \frac{27(1024\ell-99)z^4}{8192\ell+864} + \frac{243\left(1310720\ell^2-211968\ell+36207\right)z^5}{640(256\ell+27)^2} \\
&\quad + \frac{729\left(671088640\ell^3-130940928\ell^2+36951552\ell-6187023\right)z^6}{2048(256\ell+27)^3} + \mathcal{O}(z^7)
\end{aligned}
\tag{2.168}
$$

The periods are ℓ-dependent combinations of ϖ_i,

$$
\varpi(\ell,\mathrm{u}) = \alpha_0(\ell)\varpi_0 + \alpha_1(\ell)\varpi_1 + \alpha_2(\ell)\varpi_2 \tag{2.169}
$$

We can fix the ℓ dependence by using the original Picard-Fuchs equations which read $\mathcal{D}_i\varpi(\ell,\mathrm{u}) = 0$

$$
\begin{aligned}
\mathcal{D}_1 &= -\mathrm{u}^4\partial_\mathrm{u}^2 - 2\mathrm{u}^3\partial_\mathrm{u} - \mathrm{u}\partial_\mathrm{u}\partial_\ell + 3\partial_\ell + 3\ell\partial_\ell^2, \\
\mathcal{D}_2 &= (\mathrm{u}+1)\mathrm{u}\left(\mathrm{u}\partial_\mathrm{u}^2 + \partial_\mathrm{u}\right) + \ell\left(4\partial_\ell - \mathrm{u}(3\mathrm{u}+4)\partial_\ell\partial_\mathrm{u} + 4\ell\partial_\ell^2\right).
\end{aligned}
\tag{2.170}
$$

The result is a set of ordinary differential equations for the coefficients $\alpha_i(\ell)$,

$$
\begin{aligned}
\ell(256\ell+27)^2\alpha_1''(\ell) + (512\ell+27)(256\ell+27)\alpha_1'(\ell) + (16384\ell+1920)\alpha_1(\ell) &= 0, \\
81\ell\alpha_0''(\ell) + 81\alpha_0'(\ell) + 24\alpha_1'(\ell) + \frac{512\alpha_1(\ell)}{256\ell+27} &= 0, \\
\alpha_2(\ell) = \frac{81\left(\left(65536\ell^2-3456\ell+243\right)\alpha_1(\ell) - \left(27648\ell^2+2916\ell\right)\alpha_1'(\ell)\right)}{64(256\ell+27)^2}
\end{aligned}
\tag{2.171}
$$

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